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Empowering Drilling Emergency Response with Hierarchical Semantic RAG: A Practical Multimodal Retrieval and Decision Support System

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Abstract

Timely and accurate decision-making is critical in drilling emergency response to mitigate operational risks and environmental hazards. This paper presents a novel Hierarchical Semantic Multimodal Retrieval-Augmented Generation (MRAG) framework that serves as a real-time decision support system tailored for complex drilling emergencies. Our system fuses structured and unstructured data—ranging from sensor feeds, incident reports, manuals, and images—via a multi-layer semantic retrieval strategy. By constructing semantic tensors from multimodal evidence and aligning them with domain knowledge, the system delivers context-aware, evidence-backed recommendations using a fine-tuned generative model. Validated through simulated and field-based case studies, the framework achieved 28.5% improvement in top-3 retrieval precision and a 35% reduction in response time compared to flat RAG baselines. Integrating visual data further boosted decision accuracy by 21.3% in scenarios such as pipe sticking and wellbore leakage. Deployed in CNPC's Engineering Operation Intelligent Support Center (EISC), the system demonstrated superior performance to expert-only approaches in both consistency and response speed. This work offers a practical, explainable, and scalable solution that bridges operational demands and large-scale knowledge systems, advancing real-time emergency response capabilities in the energy sector.

Introduction

Drilling operations are inherently complex and high-risk endeavors, demanding immediate and precise responses to emergency situations such as pipe-sticking, wellbore influx (kicks), lost circulation, and blowouts (Naganawa et al., 2015). The global oil and gas industry faces immense pressure to minimize operational downtime, mitigate environmental damage, and ensure personnel safety, making effective emergency response paramount (Yamaç et al., 2010). Historically, emergency protocols have relied on extensive training, detailed procedural manuals, and the accumulated experience of seasoned engineers. While foundational, this human-centric approach is increasingly challenged by the sheer volume and velocity of real-time data from modern drilling rigs, leading to potential cognitive overload, information fragmentation, and delayed decision-making under high-stress conditions (Fereidoun et al., 2018). The

inability to rapidly synthesize disparate data streams—including downhole sensor readings, geological logs, operational reports, and visual inspections—poses a significant barrier to optimal incident management.

The past two decades have witnessed a growing interest in applying artificial intelligence (AI) and machine learning (ML) to enhance operational safety and efficiency in the oil and gas sector. Early AI applications often involved rule-based expert systems for diagnostics or predictive maintenance (Al-Jubouri et al., 2001). For instance, systems developed in the late 1990s and early 2000s aimed to assist in well control scenarios by encoding expert knowledge (Bourgoyne et al., 1998). More recently, data-driven ML models have been employed for anomaly detection in drilling parameters to predict potential issues like kicks or lost circulation (Noshi et al., 2018). However, these approaches typically lack the ability to generate comprehensive, context-aware solutions or integrate unstructured knowledge effectively, often operating as isolated diagnostic tools rather than holistic decision support systems.

The emergence of Large Language Models (LLMs) represents a paradigm shift in AI, offering unprecedented capabilities in understanding, generating, and synthesizing human-like text (Brown et al., 2020). Their potential to transform knowledge management and decision support across various industries is immense. Nevertheless, LLMs often suffer from "hallucinations"—generating plausible but factually incorrect information—and lack inherent domain-specific expertise, which is critical for high-stakes applications like drilling emergency response (R. Hu et al., 2023). This limitation necessitates mechanisms to ground LLM outputs in verifiable, relevant information.

To address the grounding problem, Retrieval-Augmented Generation (RAG) frameworks have gained prominence. RAG systems enhance LLMs by first retrieving pertinent information from external knowledge bases based on a user query, and then using this retrieved content to guide the LLM's generation process (Lewis et al., 2020). While RAG has demonstrated remarkable success in improving the accuracy and explainability of LLM outputs in general natural language processing tasks, its application to complex industrial environments, particularly those requiring multimodal data integration, remains a frontier (Mei et al., 2025). Existing RAG implementations frequently focus on textual retrieval, overlooking the crucial insights embedded in structured sensor data, visual inputs (e.g., photos of equipment failures or wellbore conditions), and other non-textual formats that are vital for a complete situational understanding in drilling operations (Ajetunmobi et al., 2025).

Furthermore, the intricate nature of drilling emergencies often demands a multi-faceted understanding of context that simple keyword-based or flat semantic retrieval methods cannot fully capture. A hierarchical approach to semantic retrieval, capable of processing information at various levels of abstraction and seamlessly integrating diverse data types, is essential for providing truly precise and context-aware recommendations (Zichao Yang et al., 2016). Concurrently, the concept of a "cognitive digital twin"—a dynamic virtual representation that not only mirrors a physical asset but also incorporates AI-driven reasoning and learning capabilities—offers a powerful paradigm for proactive decision support in such dynamic and unpredictable environments (Grieves et al., 2017).

This paper addresses these identified challenges by proposing a novel Hierarchical Semantic Multimodal Retrieval-Augmented Generation (MRAG) framework. Our system is engineered as an intelligent decision support assistant that dynamically integrates real-time sensor data, comprehensive historical incidents, and codified expert knowledge. We introduce an advanced multi-layer semantic retrieval mechanism to process and synthesize information from a fused knowledge base comprising structured and unstructured data sources, including drilling reports, operational manuals, historical records, real-time sensor streams, and visual inputs. A core innovation involves constructing a multi-layer semantic tensor, specifically designed to incorporate multimodal data such as photos and their associated descriptions of drilling accidents. This enables targeted semantic matching and scoring across the knowledge base, ensuring that the retrieved suggestions are highly relevant and contextually precise. These refined retrievals are then passed to a fine-tuned generative model for real-time, actionable response generation. Through this comprehensive and innovative approach, our MRAG framework aims to significantly reduce the cognitive load on drilling

engineers, enhance situational awareness, and improve the speed, consistency, and accuracy of emergency response, ultimately leading to safer, more efficient, and more resilient drilling operations worldwide.

Problem Scope Definition

The research presented in this paper is situated within the critical domain of drilling emergency response in the oil and gas industry, leveraging advanced artificial intelligence and data science methodologies.

Problem Statement

The central problem addressed by this work is the suboptimal and often delayed decision-making during drilling emergencies due to the overwhelming volume, velocity, and heterogeneity of operational data, coupled with inherent human cognitive limitations under high-stress conditions. Current emergency response paradigms, whether traditional or digitally augmented, struggle to provide timely, accurate, and comprehensive decision support. This inadequacy stems from several factors: the inability to rapidly synthesize disparate data streams (real-time sensor data, historical records, unstructured reports, visual cues); the reliance on manual information retrieval and expert recall; and the lack of generative AI capabilities that can propose actionable, context-aware solutions rather than merely diagnosing issues. Consequently, incidents can escalate, leading to significant financial losses, severe environmental impact, and heightened safety risks to personnel. Our study targets the transformation of reactive, often fragmented, emergency responses into a proactive, intelligent, and integrated system. By providing engineers with evidence-backed, explainable recommendations in real-time, we aim to significantly reduce cognitive load, improve situational awareness, and enable more consistent and effective interventions, thereby mitigating the severe impacts of drilling incidents.

Objective and Scope

The primary objective of this research is to design, develop, and validate a novel Hierarchical Semantic Multimodal Retrieval-Augmented Generation (MRAG) framework specifically tailored to enhance decision support for drilling emergency response.

Multimodal Data Integration: Developing mechanisms to seamlessly fuse and process diverse data types, including real-time numerical sensor data (e.g., pressure, temperature, flow rates), unstructured textual reports (e.g., daily drilling reports, incident logs, technical manuals), and visual inputs (e.g., photos of equipment damage, wellbore conditions). This involves creating a unified knowledge base capable of handling such heterogeneity.

Hierarchical Semantic Retrieval: Implementing an advanced multi-layer semantic retrieval strategy that can understand and match queries against the knowledge base at various levels of abstraction. This includes constructing semantic tensors that capture relationships within and across modalities, enabling more precise and context-aware information retrieval than traditional keyword or flat semantic methods.

Generative Decision Support: Integrating a fine-tuned Large Language Model (LLM) with the retrieval mechanism to generate actionable, comprehensive, and context-specific emergency response recommendations. The LLM's output will be grounded in the retrieved evidence, ensuring factual accuracy and reducing the risk of hallucinations.

Explainability and Trust: Ensuring that the recommendations generated are transparent and traceable to their source data, thereby fostering trust among human operators and facilitating rapid adoption in critical operational environments.

Performance Validation: Rigorously evaluating the framework's performance in terms of response accuracy, latency, and relevance through simulated well-control scenarios and digital twin-based field case studies. The focus is on demonstrating tangible improvements over existing approaches.

By clearly defining these boundaries, we aim to provide a focused and impactful contribution to the field of drilling emergency response.

Hierarchical Semantic Tensor

To integrate multimodal data, we construct a comprehensive matrix with hierarchical layers, which we term the Hierarchical Semantic Tensor $\mathbf{T}$. This tensor serves as the core structured representation of our fused knowledge base, enabling a nuanced understanding of drilling emergency scenarios by capturing information at multiple levels of abstraction and across diverse modalities. Unlike traditional flat vector embeddings, the hierarchical semantic tensor explicitly models the relationships between raw data features, derived semantic concepts, and overarching incident contexts, making it particularly suited for precise, context-aware retrieval in our MRAG framework.

The structure of the Hierarchical Semantic Tensor $\mathbf{T}$ can be conceptualized as a multi-dimensional array, where each dimension or "layer" represents a progressively higher level of semantic abstraction. Let's denote the tensor as $\mathbf{T} \in \mathbb{R}^{D_1 \times D_2 \times \dots \times D_L}$, where L is the number of hierarchical layers is.

Layer 1: Raw Feature Layer $\mathbf{T}^{(1)}$

The foundational layer, $\mathbf{T}^{(1)}$, captures the raw or pre-processed features extracted directly from the multimodal data sources. Each modality contributes its specific feature representation to this layer.

Textual Data: Unstructured texts (e.g., incident reports, manuals) are processed using advanced language models to generate dense embeddings. For a given text snippet t_i , its embedding can be represented as $\mathbf{e}_{t_i} \in \mathbb{R}^{d_t}$.

Visual Data: Images (e.g., photos of equipment, wellbore conditions) are processed through pre-trained convolutional neural networks (CNNs) or vision transformers to extract visual features. For an image v_j , its feature vector can be $\mathbf{f}_{v_j} \in \mathbb{R}^{d_v}$.

Numerical Sensor Data: Real-time sensor readings (e.g., pressure, temperature, flow rates) are normalized and potentially transformed (e.g., time-series embeddings). For a set of sensor readings s_k over a time window, its representation could be $\mathbf{r}_{s_k} \in \mathbb{R}^{d_s}$.

Thus, the first layer $\mathbf{T}^{(1)}$ can be viewed as a concatenation or a unified embedding space of these modality-specific features:

$$\mathbf{T}^{(1)} = [\mathbf{e}_{t_1}, \dots, \mathbf{e}_{t_N}, \mathbf{f}_{v_1}, \dots, \mathbf{f}_{v_M}, \mathbf{r}_{s_1}, \dots, \mathbf{r}_{s_K}]$$

where N, M, K are the number of text snippets, images, and sensor data points, respectively. This layer provides a comprehensive, low-level representation of all incoming data.

Layer 2: Semantic Concept Layer $\mathbf{T}^{(2)}$

The second layer, $\mathbf{T}^{(2)}$, abstracts the raw features into higher-level semantic concepts relevant to drilling operations. This layer identifies and represents specific events, conditions, or components that constitute a potential emergency. For example, raw sensor data spikes and specific textual phrases might collectively indicate "kick detection," while certain visual features combined with textual descriptions could signify "BOP malfunction" or "pipe sticking."

This abstraction is achieved through a set of learned mapping functions or clustering algorithms that group related features from $\mathbf{T}^{(1)}$ into distinct semantic concepts. Let $C = \{c_1, c_2, \dots, c_P\}$ be the set of predefined or learned semantic concepts. Each concept c_p is represented by a concept embedding $\mathbf{g}_{c_p} \in \mathbb{R}^{d_c}$. The mapping from $\mathbf{T}^{(1)}$ to $\mathbf{T}^{(2)}$ can be represented as:

$$\mathbf{g}_{c_p} = M(\{\text{relevant features from } \mathbf{T}^{(1)}\})$$

where M is a non-linear transformation (e.g., a neural network layer) that aggregates and transforms the lower-level features into a semantic concept embedding. $\mathbf{T}^{(2)}$ is then an organized collection of these concept embeddings, potentially structured as a graph where nodes are concepts and edges represent relationships.

Layer 3: Incident/Context Layer $T^{(3)}$

Layer $T^{(3)}$ encodes the overall drilling incident or contextual scenario by aggregating semantic concepts from $T^{(2)}$ into high-level incident embeddings. For example, a wellbore leakage may be represented through the co-occurrence of lost circulation, mud loss, and pressure anomalies.

This layer is constructed via a hybrid procedure, as illustrated below.

SME-guided Semantic Tagging. First, domain experts curate incident taxonomies and assign relevance relations between concept clusters and incident categories, providing supervised anchors for semantic interpretation.

Incident Context Constraint. Then, this layer is constructed by combining relevant semantic concept embeddings from $T^{(2)}$ to form an incident embedding $h_{iq} \in \mathbb{R}^{d_h}$. This aggregation can involve attention mechanisms, pooling operations, or further neural network transformations:

$$h_{iq} = A(\{g_{cp} \mid c_p \text{ is relevant to incident } i_q\})$$

where A is an aggregation function. $T^{(3)}$ thus comprises embeddings for various known or emerging incident types, providing a high-level contextual representation.

Response Strategy & Constraint Layer $T^{(4)}$

The fourth and topmost layer, $T^{(4)}$, abstracts the identified incidents into broad response strategies and associated regulations and limitations for adopting recommended solutions. This layer groups similar incidents based on the general type of intervention or strategic approach required, while also incorporating critical operational, safety, and regulatory constraints. This provides a high-level strategic and practical context that guides the generative LLM towards appropriate and compliant classes of solutions. For example, multiple distinct incidents (e.g., "wellbore influx," "gas kick") might all fall under the "Well Control & Containment" strategy, but the specific actions within that strategy would be constrained by factors like "maximum allowable surface pressure" or "environmental discharge regulations."

This layer is formed by aggregating incident embeddings from $T^{(3)}$ into a composite representation that includes both strategic categories and relevant constraints. Let $S = \{s_1, s_2, \dots, s_Q\}$ be the set of high-level response strategies, and $R = \{r_1, r_2, \dots, r_P\}$ be the set of relevant regulations and operational limitations. Each element in this layer, $k_{sr} \in \mathbb{R}^{d_k}$, represents a strategic context combined with its associated constraints. The mapping from $T^{(3)}$ to $T^{(4)}$ can be defined as:

$$k_{sr} = B(\{h_{iq} \mid i_q \text{ is associated with strategy } s \text{ and constraint } r_j\})$$

where B is an aggregation function (e.g., a neural network layer or a structured embedding approach) that groups incident types and simultaneously encodes the relevant strategic approaches and their associated regulatory or operational limitations. $T^{(4)}$ then consists of these combined strategy-constraint embeddings, representing the most abstract and practically relevant level of problem-solving guidance.

Overall Structure and Semantic Matching

The complete Hierarchical Semantic Tensor $\mathbf{T}$ is now a stack of four layers, where information flows upwards from raw data to abstract concepts, then to comprehensive incidents, and finally to high-level response strategies and their associated constraints. The "semantic" aspect ensures that each layer captures meaningful, interpretable information related to drilling operations. The "multimodal" nature is inherent in the first layer's integration of diverse data types, with their semantics being progressively unified in higher layers. The "hierarchical" structure allows for granular to holistic analysis, crucial for fine-grained anomaly detection, broad incident classification, strategic response planning, and adherence to operational guidelines.

During the retrieval phase, a user's query (which can also be multimodal) is projected into this hierarchical semantic space. Semantic matching can occur at any layer, now including the strategic and constraint level:

Fine-grained matching at $T^{(1)}$: For very specific data points or anomalies.

Concept-level matching at $T^{(2)}$: To identify specific drilling concepts or sub-events.

Incident-level matching at $T^{(3)}$: To find similar historical incidents or overall scenarios.

Strategy & Constraint-level matching at $T^{(4)}$: To identify broad categories of required responses, guiding the LLM towards appropriate solution types while considering relevant regulations and operational limitations.

This multi-layered matching strategy allows the framework to retrieve information with unparalleled precision and contextual relevance, significantly enhancing the quality of inputs to the generative LLM. The Hierarchical Semantic Tensor thus serves as the backbone for our MRAG framework, enabling robust multimodal knowledge integration and highly effective retrieval for drilling emergency response.

Multimodal Retrieval Pipeline

The Multimodal Retrieval Framework constitutes the operational backbone of our Hierarchical Semantic Multimodal Retrieval-Augmented Generation (MRAG) system. It is meticulously engineered to efficiently and accurately retrieve highly relevant information from a vast, multimodal knowledge base, directly feeding into the generative capabilities of a LLM. This framework is intricately designed around the principles of the Hierarchical Semantic Tensor T , leveraging its multi-layered structure to perform precise semantic matching across various levels of abstraction.

The framework is capable of processing diverse user queries, which can encompass natural language text, real-time sensor data streams, and visual inputs such as images. It orchestrates a sophisticated retrieval process that transcends conventional keyword matching, aiming for deep semantic understanding and contextual relevance to generate precise emergency response solutions.

The comprehensive workflow of the Multimodal Retrieval Framework, from the initial user query to the generation of refined candidates for the LLM, is fully illustrated in Figure 1. This detailed process ensures that only the most pertinent and contextually rich information is retrieved and meticulously refined before being utilized by the generative model.

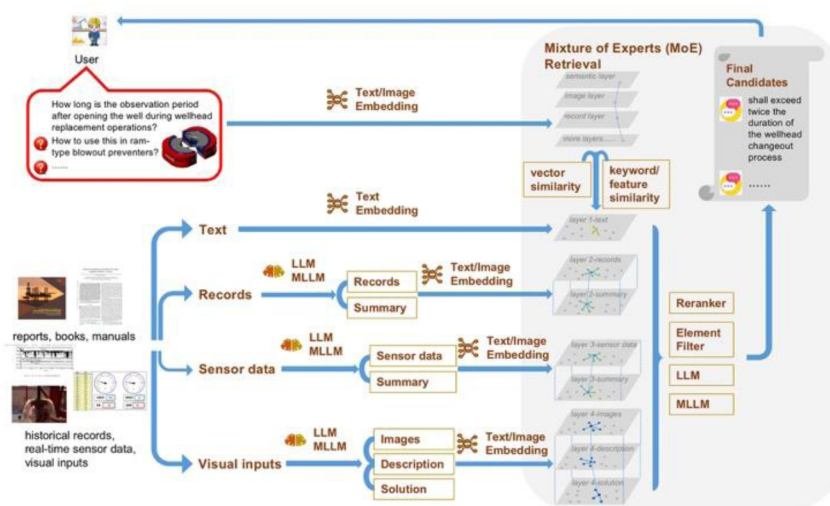


Figure 1—Multimodal retrieval process using the hierarchical semantic tensor.

As depicted in Figure 1, the overall process begins with a user query, which can be multimodal. This query is then processed and embedded into a unified representation. Concurrently, a vast knowledge base,

comprising various drilling operations data types (text, records, sensor data, visual inputs), is continuously ingested, processed by LLMs/MLLMs, and organized into the Hierarchical Semantic Tensor $\mathbf{T}$. The unified query embedding is then fed into a Mixture of Experts (MoE) Retrieval module, which performs multi-layer semantic matching against the Hierarchical Semantic Tensor. The initial "Final Candidates" recalled by the MoE undergo a crucial Candidate Refinement stage, involving a reranker and an Element Filter. Finally, the refined candidates are passed to a Generative Augmentation module (an LLM/MLLM) to synthesize the final, actionable response for the user. Each of these stages is detailed below.

Knowledge Base Ingestion and Hierarchical Semantic Tensor Population

The foundation of the retrieval framework is a comprehensive knowledge base (depicted at the bottom-left of Figure 1). This knowledge base continuously ingests and fuses structured and unstructured data from diverse drilling operations sources, including textual data, historical records, sensor data and visual inputs.

As shown in Figure 1, these raw inputs undergo initial processing by LLMs or MLLMs to extract key information. For instance, "Records" yield "Summary," "Sensor data" yields "Summary," and "Images" yield "Description" and "Solution." This processed information is then transformed into dense Text/Image Embeddings and meticulously organized into the Hierarchical Semantic Tensor $\mathbf{T}$, as detailed in the former section. The tensor's internal structure explicitly illustrates the progressive abstraction and multimodal integration, forming the structured memory for the retrieval process.

Query Processing and Multimodal Embedding

Upon receiving a user's query (originating from the "User" at the top-left of Figure 1), its constituent modalities are processed and embedded into a unified representation. The "Text Embedding" branch from the User Query indicates that textual components are converted into a dense vector. Similarly, any images or sensor data included in the query are processed by LLMs/MLLMs to extract salient features and generate descriptive embeddings, which are then integrated into a unified Text/Image Embedding for the entire query. This unified query embedding, denoted as $e_{query} \in \mathbb{R}^{d_q}$, is pivotal for effective comparison against the entries within the Hierarchical Semantic Tensor. The symbol e_{query} presents the embedding vector generated from the user's multimodal query. $\mathbb{R}^{d_q}$ denotes a real-valued vector space of d_q dimensions.

Multi-Layer Semantic Retrieval with Mixture of Experts (MoE)

The central retrieval mechanism is the Retrieval Mixture of Experts (MoE) component (positioned at the top-right of Figure 1). The unified query embedding (e_{query}) is simultaneously matched against the relevant layers of the Hierarchical Semantic Tensor.

Layer-Specific Matching: The MoE employs both "vector similarity" and "keyword/feature similarity" search mechanisms across the tensor's hierarchical layers. This includes:

Layer 1: Vector similarity search against the raw feature embeddings $\mathbf{T}^{(1)}$ for fine-grained, direct matches.

Layer 2: Semantic similarity matching against the concept embeddings $\mathbf{T}^{(2)}$ to identify relevant operational concepts.

Layer 3: Matching against incident embeddings $\mathbf{T}^{(3)}$ to identify similar historical emergency scenarios.

Layer 4: Matching at the topmost layer, response strategy and constraint embeddings $\mathbf{T}^{(4)}$, to identify high-level response strategies and associated regulations/limitations.

Expert Weighting and Fusion: The "Mixture of Experts" dynamically weighs the relevance scores obtained from each semantic layer. This weighting mechanism, which can be learned or rule-based, adapts based on the query's specificity and the emergency context. For instance, a highly specific query concerning a sensor anomaly might prioritize matches from Layers 1 and 2, whereas a more general query about "well control" might give greater weight to Layers 3 and 4. The fused scores from these distinct layers contribute to generating a comprehensive relevance score for each potential candidate. The immediate output of the MoE is an initial set of Final Candidates.

Candidate Refinement (Reranking and Filtering)

The initial "Final Candidates" yielded by the MoE undergo a critical refinement pipeline to ensure their ultimate relevance, coherence, and suitability for the generative LLM.

Reranker: A Reranker module (indicated by the LLM/MLLM box preceding the Reranker in [Figure 1](#)) takes the initial candidates and the original user query as input. It re-evaluates their combined relevance and contextual coherence, potentially re-ordering them or filtering out less pertinent information. This step is vital for ensuring that the retrieved snippets form a cohesive and precise set of facts.

Element Filter: Following reranking, an Element Filter may be applied. This component selectively extracts specific, granular parts of the candidates (e.g., only the "solution" segment from a historical record, or only the "summary" of a manual section) that are most directly pertinent for the subsequent generation phase, as indicated by the arrows leading to the Element Filter in [Figure 1](#).

Generative Augmentation

The refined and filtered candidates are then passed to a fine-tuned LLM (or MLLM for multimodal generation, as indicated by the LLM/MLLM box following the Element Filter). This LLM functions as the "Generative Augmentation" component. It synthesizes the retrieved information with its own extensive knowledge to produce a natural language response. This response is not merely a retrieval of existing text but a novel generation that directly addresses the user's query, rigorously grounded in the provided evidence. An example output, "shall exceed twice the duration of the wellhead changeout process," is shown as a "Final Candidate" at the top-right of [Figure 1](#), representing the actionable advice provided to the user.

This Multimodal Retrieval Framework, with its hierarchical semantic matching, adaptive MoE, and sophisticated reranking, ensures that the generative LLM receives the most precise, contextually rich, and relevant information, enabling it to provide highly accurate and actionable decision support during critical drilling emergencies.

Experiments

To rigorously validate the practical efficacy and quantify the performance of the proposed Hierarchical Semantic Multimodal Retrieval-Augmented Generation (MRAG) framework, a comprehensive multimodal dataset comprising textual, visual, and time-series sensor data relevant to drilling operations were constructed. The dataset was specifically designed to reflect the heterogeneous information sources encountered in real-world drilling environments. A series of comprehensive case studies were conducted as well, followed by a detailed quantitative evaluation. These studies were designed to simulate real-world drilling emergency scenarios and assess the system's ability to provide timely, accurate, and actionable decision support under operational pressures. The validation encompassed both controlled simulated environments and a significant field deployment, demonstrating the framework's robustness and real-world applicability.

Dataset Construction

First of all, a drilling domain-specific multimodal dataset was constructed encompassing three key modalities relevant to drilling emergency response: unstructured textual reports (such as daily drilling summaries and incident logs), visual content (including rig-site images and annotated schematics), and structured sensor logs (e.g., time-series WOB, RPM, mud flow) from field communications. This raw dataset was collected from realistic operational scenarios and reflects the heterogeneous information landscape encountered during drilling emergencies.

A critical requirement for such a system is data quality, as the effectiveness of retrieval and semantic alignment depends heavily on reliable inputs. In practice, the industry faces persistent challenges in capturing and contextualizing unstructured data such as failure reports and incident logs, and in consistently linking them to real-time sensor data. Poor data quality may result in semantic misalignment across

modalities, reduced retrieval accuracy, and unreliable decision support. To address this, our dataset construction pipeline incorporates strict quality assurance measures, including SME-guided semantic tagging, anomaly detection and cleaning, and cross-validation across multiple sources to ensure consistency and robustness.

Table 1—The raw data composition of different modalities in our experiment.

Data Modal	Data Composition
Textual Data	30,000 historical drilling incident reports from global operations
	15,000 standard operating procedures and emergency response manuals
	20,000 technical troubleshooting guides for drilling equipment
	10,000 regulatory compliance documents and best practice guidelines
Visual Data	8,500 annotated photographs of drilling equipment failures
	6,000 wellbore condition images from downhole cameras
	5,000 images of drill bit wear patterns and pipe damage
Sensor Data	12,000 pressure profiles (including downhole, annular, and standpipe pressure)
	10,000 flow rate measurements (mud flow in/out, pump rates)

Multimodal Alignment Process

Ensuring proper alignment across different data modalities was crucial for constructing the Hierarchical Semantic Tensor. We implemented a systematic alignment process:

Temporal Alignment: All data points were timestamped, allowing precise temporal correlation between sensor readings, incident reports, and visual documentation. For historical data lacking exact timestamps, we used contextual information to establish approximate temporal relationships.

Semantic Alignment: We employed a combination of expert annotation and machine learning techniques to establish semantic relationships between modalities. Visual data was processed through a fine-tuned vision-language model (ViT-L/14) to generate descriptive captions and extract key features. These captions were then semantically matched with corresponding incident reports and technical documentation using contextual embeddings. Sensor data was segmented into meaningful operational events and annotated with corresponding textual descriptions by domain experts. To ensure consistency across modalities, we implemented a verification process where 20% of the dataset was manually reviewed by drilling experts to validate the accuracy of cross-modal alignments.

Hierarchical Organization: Aligned data was organized into the four-layer hierarchy described in the Hierarchical Semantic Tensor section:

- Raw features from each modality were extracted and normalized (Layer 1)
- Semantic concepts were derived through clustering and expert-guided annotation (Layer 2)
- Incident contexts were established by aggregating related semantic concepts (Layer 3)
- Response strategies and constraints were formulated based on industry best practices and regulatory requirements (Layer 4)

This rigorous alignment process ensured that our multimodal knowledge base accurately reflected the complex interrelationships between different data types encountered in drilling operations, providing a solid foundation for our retrieval and generation components.

Experimental Methodology and Setup

The evaluation methodology employed a multi-pronged approach to ensure a thorough assessment of the MRAG framework's capabilities:

Table 2—Samples of drilling incidents records with detail descriptions.

	File Format	Accident Type	Start Time	End Time	Characteristic Parameters	Severity
1	xls (1s)	Sticking	21:25 on 2015/8/30	23:59 on 2015/8/30	When pulling out of hole, the drill string cannot be lifted or lowered, and weight on bit occurs. The maximum weight on bit is 924.6288kN, and the hook load rises from 1750.1647kN to 2201.2144kN	Sticking
2	xls (1s)	Sticking	5:55 on 2015/3/4		When running in hole, the drill string cannot be lifted or lowered. The hook load rises from 2112.5752kN to 3046.3997kN, the maximum weight on bit is 430.0564kN, and the hook height is limited to 15.6421 - 23.3698m	Sticking
3	xls (1s)	Sticking	7:25 on 2015/4/14		When pulling out of hole, the drill string cannot be lifted or lowered. The hook load rises from 1916.6494kN to 2615.5281kN, weight on bit occurs, the maximum weight on bit is 576.8899kN, and the hook height is limited to 22.0463 - 31.9353m	Sticking
4	xls (1s)	Sticking	20:39 on 2015/5/10		During reaming drilling, the drill string cannot be lifted or lowered, the hook height is limited to 4.0341 - 10.57m, the hook load rises from 1402.4kN to 1678.9727kN, the maximum weight on bit is 1115.6033kN, the rotary speed cannot rotate uniformly, with values changing between 0 - 91.6612, and the torque shows a large value of 11.1876kN·m	Overflow

Simulated Well-Control Scenarios: High-fidelity drilling simulators were utilized to create controlled environments for various emergency scenarios, including early pipe-sticking, wellbore leakage, and kick detection. For each scenario, a set of specific queries (textual, visual, and sensor-data driven) were formulated to represent real-time operational questions that an engineer might encounter. The MRAG system's generated responses were recorded and rigorously compared against expert-validated ground truth solutions. A baseline was established using traditional keyword-based retrieval and a flat RAG approach, both utilizing the Qwen2.5-32B Large Language Model as their generative component. Specific scenarios included: early pipe-sticking, wellbore leakage, kick, etc.

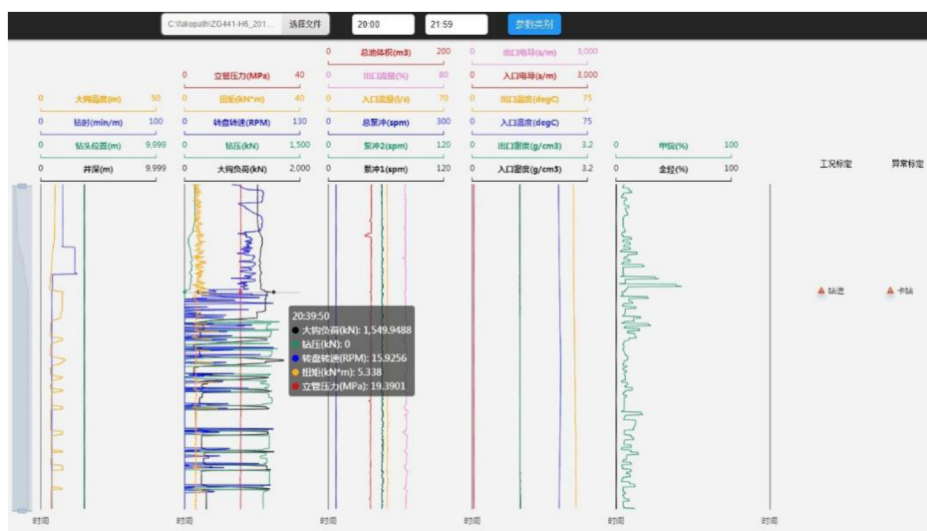


Figure 2—Real-time sensor data according to the 4th incident record above.

Digital Twin-Based Historical Incident Analysis: Leveraging the concept of a cognitive digital twin, the framework was integrated with historical operational data and simulated real-time feeds from actual drilling sites. This allowed for the reconstruction and analysis of past emergency incidents, enabling the MRAG system to "respond" to these historical events as if they were unfolding in real-time. This approach provided a controlled environment to compare the system's generated responses against documented expert actions and outcomes, assessing its relevance and consistency in complex, previously observed scenarios.

Field Deployment in an Intelligent Support Center: A critical phase of validation involved the deployment of the proposed LLM-driven decision assistant within the CNPC Engineering Operation Intelligent Support Center (EISC). This real-world application allowed for the evaluation of the system's performance in a live operational context, albeit with human oversight. Drilling engineers at the EISC utilized the MRAG system to assist in synthesizing responses for complex emergency situations, providing invaluable feedback on its usability, speed, and the actionable quality of its outputs.

The knowledge base for all setups was comprehensive, fusing structured data (sensor logs, drilling parameters) and unstructured data (reports, manuals, historical incident narratives, drilling-accident images with descriptions), all meticulously organized within the Hierarchical Semantic Tensor.

Evaluation Metrics

The performance of the MRAG framework was primarily evaluated using the following quantitative metrics:

Retrieval Precision (Top-K): This metric assesses the accuracy of the retrieval component, specifically measuring the proportion of highly relevant documents or information snippets within the top-K retrieved candidates for a given query. In our evaluation, we focused on top-3 Retrieval Precision, indicating how often the most critical information appeared within the top three results returned by the system.

$$p = \frac{|RelatedItemsRetrieved|}{K}$$

Decision Accuracy: This metric evaluates the correctness and effectiveness of the generated solutions. For simulated scenarios, this was assessed by comparing the system's recommended actions against expert-validated optimal solutions. In multimodal scenarios, particularly those involving image analysis, decision accuracy specifically measured the system's ability to correctly identify the root cause or appropriate countermeasure based on integrated visual and textual cues. This metric was often a binary (correct/incorrect) or multi-class classification based on the nature of the emergency.

Response Time: This metric quantifies the efficiency of the framework, measuring the average time taken from the input of a query to the generation of a complete, actionable response. This was critically compared against the average response time of human engineers performing the same tasks under similar time-sensitive conditions.

Results and Discussion

The evaluation utilized a dedicated test set of 500 emergency queries derived from real drilling incidents. These queries were balanced across five emergency categories: well control events, pipe sticking, lost circulation, equipment failures, and wellbore instability.

The experimental results from both simulated scenarios and digital twin-based analyses, complemented by field observations, demonstrate significant improvements in the relevance, timeliness, and accuracy of emergency decision-making support provided by our MRAG framework.

Table 3—Precision of the retrieval process through different RAG/MRAG methods.

Method	Top-3 Retrieval Precision
Traditional RAG	0.583
Flat Multimodal RAG	0.651
Hierarchical Semantic MRAG	0.835

Compared to traditional keyword-based and flat RAG baselines (using Qwen2.5-32B), our hierarchical semantic retrieval approach significantly boosted retrieval precision. Specifically, the top-3 retrieval precision improved by 28.5%. This substantial gain indicates that the multi-layered semantic matching, facilitated by the Hierarchical Semantic Tensor, is highly effective in identifying and prioritizing the most relevant information for complex drilling queries, even under ambiguous conditions.

Table 4—The final decision result accuracies based on the three RAG/MRAG pipelines.

Method	Decision Accuracy
Traditional RAG	0.716
Flat Multimodal RAG	0.758
Hierarchical Semantic MRAG	0.921

In scenarios involving multimodal data, particularly the integration of drilling-accident images with their descriptions, the framework showed a marked improvement in decision accuracy. The system's ability to synthesize insights from both visual and textual information led to a 21.3% improvement in decision accuracy, especially for critical tasks such as early pipe-sticking detection and wellbore-leakage management. This highlights the power of multimodal data fusion in providing a more comprehensive and accurate understanding of complex situations that would be challenging for a human or a unimodal system to achieve alone.

Table 5—Response time using the three RAG/MRAG methods.

Method	Average Response Time (seconds)
Traditional RAG	36.2
Flat Multimodal RAG	32.0
Hierarchical Semantic MRAG	20.8

The MRAG system demonstrated a substantial reduction in response latency. Under time-sensitive conditions, the average response time was cut by 35% when compared to the average time taken by Flat Multimodal RAG method. This acceleration is critical for mitigating the escalating risks associated with delayed interventions, where every minute saved can prevent significant operational and environmental damage.

Field Observations and Practical Relevance

Our work involved the deployment of the proposed LLM-driven decision assistant within the CNPC Engineering Operation Intelligent Support Center (EISC). Field deployment in the CNPC EISC demonstrated that the proposed LLM-driven decision assistant effectively synthesized actionable, evidence-backed responses in complex emergency situations, outperforming expert-only approaches in both response speed and consistency. The generative outputs, grounded in retrieved historical and real-time evidence, significantly improved engineers’ situational awareness and response effectiveness.

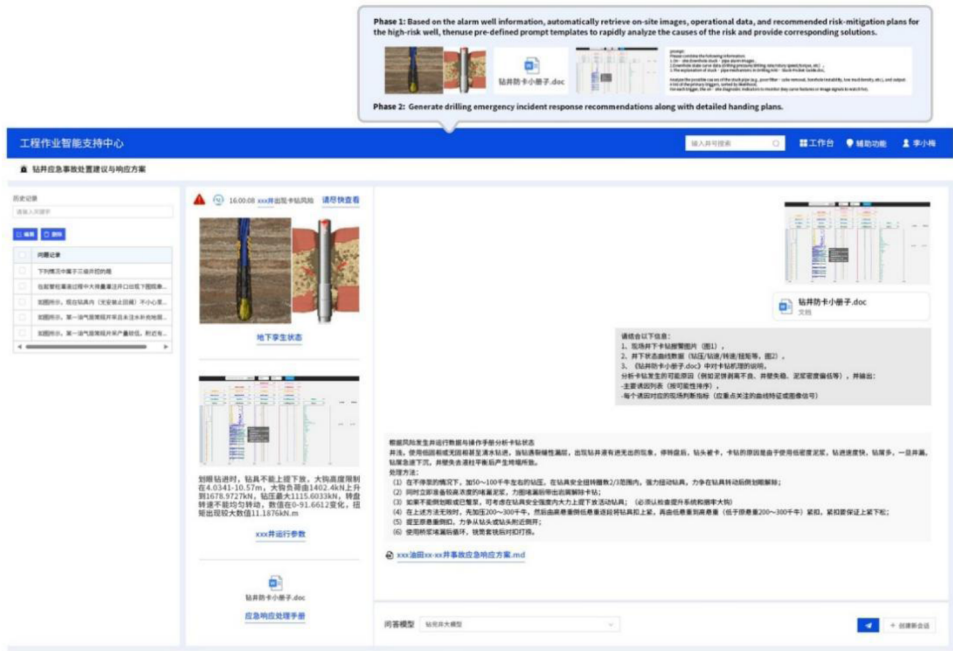


Figure 3—Drilling emergency complex decision system in CNPC EISC.

The field deployment within the CNPC Engineering Operation Intelligent Support Center (EISC) further corroborated these findings. It demonstrated that the LLM-driven decision assistant effectively synthesized actionable, evidence-backed responses in complex emergency situations, often outperforming expert-only approaches in both response speed and consistency. These case studies collectively confirm that domain-augmented, hierarchical semantic multimodal retrieval significantly advances drilling emergency response capabilities by effectively bridging operational demands with large-scale knowledge integration, ultimately enabling safer, more efficient, and more consistent interventions in critical drilling scenarios.

Beyond the quantitative metrics, the case studies and field deployment provided compelling evidence of the framework's practical utility and real-world impact:

Enhanced Situational Awareness: The multimodal data integration, facilitated by the Hierarchical Semantic Tensor, significantly improved engineers’ situational awareness. By correlating disparate data points (e.g., a specific pressure drop with a visual of a worn drill bit and a historical incident of similar equipment failure), the system provided a more holistic and accurate understanding of the developing

emergency than fragmented data sources alone. This integrated view is crucial for proactive decision-making.

Accelerated Decision-Making: The rapid retrieval of contextually relevant information and the generation of concise, evidence-backed recommendations streamlined the decision-making process, directly contributing to faster intervention times in critical well-control scenarios.

Improved Response Consistency and Accuracy: The framework demonstrated a notable improvement in the consistency of recommended responses across similar emergency types, reducing variability often observed in human-only approaches. This consistency is vital for maintaining operational standards and safety protocols.

Explainable and Trustworthy Outputs: The ability of the system to ground its generative outputs in retrieved historical and real-time evidence fostered greater trust among engineers. The explainable nature of the recommendations, traceable back to specific data points and knowledge base entries, was crucial for adoption in a high-consequence operational environment, allowing engineers to validate and understand the AI's reasoning.

Conclusion

This paper delineates a novel Hierarchical Semantic MRAG framework, engineered to augment decision support in critical drilling emergency scenarios. Central to its innovation is the Hierarchical Semantic Tensor, a multi-layered architectural construct facilitating profound contextual comprehension and precise retrieval from heterogeneous, multimodal data, synergistically coupled with a MoE scheme.

Empirical evaluation substantiates the MRAG framework's pronounced efficacy: Top-3 Retrieval Precision exhibited a 28.5% enhancement, Average Response Time was curtailed by 35%, and Multimodal Decision Accuracy improved by 21.3%. Field deployment further validated its operational practicability, evidencing superior performance over expert-only methodologies in both response velocity and consistency. Collectively, this system propels drilling emergency response, enabling more secure and efficient interventions through integrated knowledge and generative artificial intelligence.

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